LAB 2

GEODATA SCAVANGER HUNT

Collect and Document Geospatial Data

One assigned source and one shared study area per group.

PRESS → OR SPACE.

© Marc Rußwurm
Licensed under **Creative Commons Attribution-NonCommercial 4.0 International** (CC BY-NC 4.0).
You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.
Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

UNIVERSITÄT BONN

Learning Outcomes

After this lab, you will be able to:

Explain what a geospatial source measures and how to interpret its values.

Not comfortable — Very comfortable

Click to respond

Export and check a usable subset for a specified area of interest.

Not comfortable — Very comfortable

Click to respond

Document and share the subset so another group can use it in Lab 3.

Not comfortable — Very comfortable

Click to respond

Lab Outline and Deliverables

During the lab

1. **Collect geospatial data.** Find your group's assigned source, crop it to the shared study area, check the result, and upload the subset to Drive.

2. **Document your work.** Add two slides: one about the source and acquisition, and one about the prepared subset.

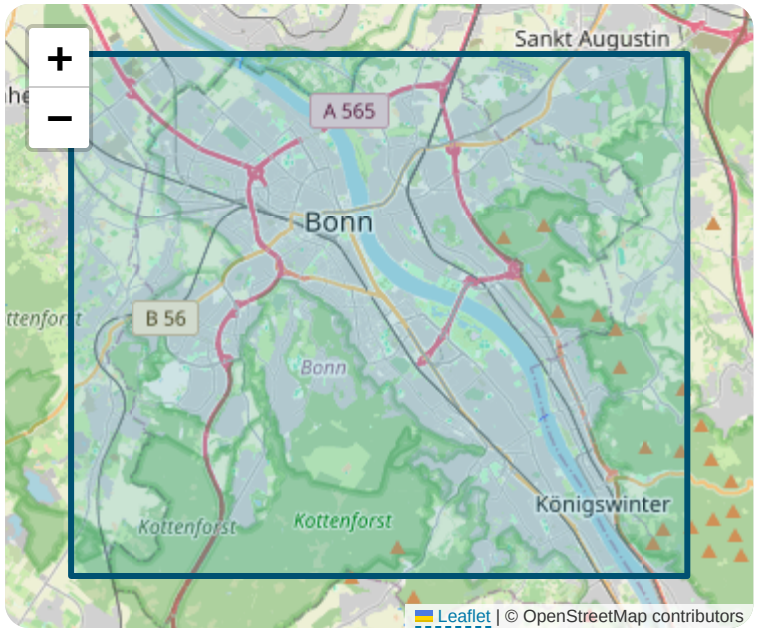
Each group submits

One usable data subset
Covering the provided map bounds and saved in the shared Google Drive folder.

Two slides in the common deck
Slide 1: source and acquisition
Slide 2: subset, metadata, and file link

In Lab 3, we will turn these files into working datasets and combine them.

Our Study Area



Geographic bounds · EPSG:4326

- W 7.010648° · E 7.222135°
- S 50.657955° · N 50.771430°

Projected corners · UTM 32N · EPSG:32632

- SW: E 359,390 · N 5,613,677 m
- SE: E 374,337 · N 5,613,297 m
- NE: E 374,640 · N 5,625,914 m
- NW: E 359,729 · N 5,626,294 m

UTM extent: E 359,390–374,640 m; N 5,613,297–5,626,294 m. Coordinates rounded to the nearest metre.

Assigned Sources · Groups 1–5

Group / assigned source	What to collect	Starting point and acquisition route
1 · **Sentinel-2 optical** (raster)	One mostly cloud-free acquisition; blue, green, red, and near-infrared bands at 10 m. Multiband GeoTIFF.	Copernicus Browser: Sentinel-2 L2A → AOI → **Analytical download** of raw bands. Alternative: Earth Engine. Browser guide.
2 · **Sentinel-1 radar** (raster)	One acquisition; VV and VH backscatter bands as GeoTIFF. Document dB units.	Sentinel-1 GRD in Earth Engine, which provides preprocessed backscatter.
3 · **Terrain elevation** (raster)	NRW DGM1 bare-earth elevation at 1 m, cropped to the AOI. GeoTIFF.	Download intersecting tiles from the DGM1 directory and crop. Product page.
4 · **Land-cover classes** (raster)	ESA WorldCover **2021** class codes at 10 m; retain the 11-class legend separately.	ESA WorldCover in Earth Engine, derived from Sentinel-1 and Sentinel-2.
5 · **ERA5-Land temperature** (raster + time)	Daily mean 2 m air temperature for **seven consecutive historical days**; dated GeoTIFFs or a documented stack.	ERA5-Land daily aggregates: select temperature_2m. This is reanalysis, not a forecast.

Assigned Sources · Groups 6–10

Group / assigned source	What to collect	Starting point and acquisition route
6 · Building footprints (polygons)	OSM building polygons and available attributes. GeoJSON.	Overpass Turbo: zoom to AOI → Wizard with building=* → run → export GeoJSON.
7 · Road and path network (lines)	OSM roads, streets, and paths with highway classification. GeoJSON line features.	Overpass Turbo: Wizard with highway=* → inspect → retain line features.
8 · Cadastral parcels (polygons)	Official Flurstücke covering the AOI. GeoJSON or zipped shapefile.	NRW geodata services: INSPIRE → Flurstücke/Grundstücke → WFS. Subset and export in QGIS.
9 · Official water features (lines / polygons)	Rivers, streams, and waterbodies intersecting the AOI; keep available feature types.	NRW geodata services: INSPIRE → Gewässernetz → Hydro – Physische Gewässer → WFS. Subset in QGIS.
10 · Nature reserves (polygons)	Naturschutzgebiete intersecting the AOI; keep names and identifiers.	NRW LINFOS downloads: download Naturschutzgebiete Shape, clip locally, export GeoJSON or a zipped shapefile.

Two Slides per Group

Slide 1 · Source and acquisition

- Explain what the source measures and which variables, units, or classes it contains.
- Give its spatial resolution and observation date or period.
- Link the original source and its documentation.
- Explain how you found, downloaded, or exported the data, including your AOI and key selections.

Slide 2 · Prepared subset

- Show a map or screenshot of the exported subset over the study area.
- Record the file format, CRS, coverage, and missing-data convention.
- Link the checked file in the shared Google Drive folder.

Add both slides to the common deck so another group can understand the source and use your file in Lab 3.